# SSH Brute Force Detection using Splunk

## Project Overview

This project demonstrates how Splunk can be used to detect SSH brute-force attacks by analysing Linux authentication logs. It focuses on identifying suspicious login attempts, attacker IP addresses, and targeted usernames.

## Introduction

Monitoring authentication logs is crucial for spotting unwanted access attempts due to the growing number of cyber threats that target network systems. The goal of this project is to identify unusual login activity and brute-force assaults by analyzing Linux SSH authentication logs using Splunk.

## Objectives

The objective of this project was to:

- Ingest and analyze Linux authentication logs in Splunk
- Detect brute-force login attempts
- Identify suspicious IP addresses and targeted usernames
- Visualize attack patterns using a dashboard

## Tools & Technologies

- **Splunk Enterprise** – Log ingestion, search, and visualization
- **Linux Authentication Logs (auth.log)** – Source of security events
- **Search Processing Language (SPL)** – Data querying and analysis

## Methodology

### Data Ingestion

A Linux authentication log file (`auth.log`) was uploaded into Splunk and indexed under the `main` index. Proper line breaking was configured to ensure each log entry was treated as a separate event.

### Data Processing and Analysis

Search queries were constructed using SPL to extract relevant security information such as IP addresses and usernames. Regular expressions were used to parse fields from raw log data.

**Detection Techniques**

The following detection strategies were implemented:

- Identification of failed login attempts
- Detection of invalid user access attempts
- Analysis of repeated authentication failures (brute-force indicators)
- Monitoring of successful SSH logins

**Key SPL Queries**

**<u>Top Attacking IPs</u>**

```
index=main ("Failed password" OR "Invalid user")
| rex "from (?<src_ip>\d+\.\d+\.\d+\.\d+)"
| stats count by src_ip
| sort -count
```

**<u>Targeted Usernames</u>**

```
index=main "Invalid user"
| rex "Invalid user (?<user>\w+)"
| stats count by user
| sort -count
```

**<u>Attack trends over time</u>**

```
index=main ("Failed password" OR "Invalid user")
| timechart count
```

**<u>Successful Logins</u>**

```
index=main "Accepted"
```

**<u>High-Risk IP Detection</u>**

```
index=main ("Failed password" OR "Invalid user" OR "maximum
authentication attempts")
| rex "from (?<src_ip>\d+\.\d+\.\d+\.\d+)"
| stats count by src_ip
| where count > 5
```

**Dashboard Development**

A consolidated Splunk dashboard was established to visualise security events. The dashboard included:

- Bar chart of top attacking IP addresses
- Visualization of targeted usernames
- Time-based trend of login attempts
- Table of successful login events
- High-risk IP detection panel

**Key Findings**

The analysis revealed several indicators of brute-force attack activity:

- Multiple IP addresses repeatedly attempted to access the system
- Commonly targeted usernames included **root, admin, support, and ftpuser**
- Certain IPs exceeded authentication thresholds, indicating automated attack behavior
- Repeated authentication failures suggested password-guessing attacks
- A successful login was observed from a recurring IP address, indicating potential authorized access or the need for further investigation

**Conclusion**

This project successfully demonstrated how Splunk can be used to detect and analyze SSH brute-force attacks. By leveraging SPL queries and visualization tools, it was possible to identify attack patterns, highlight high-risk IPs, and monitor authentication activity effectively.

The findings emphasize the importance of continuous log monitoring and proactive threat detection to secure systems against unauthorized access.

**Recommendations**

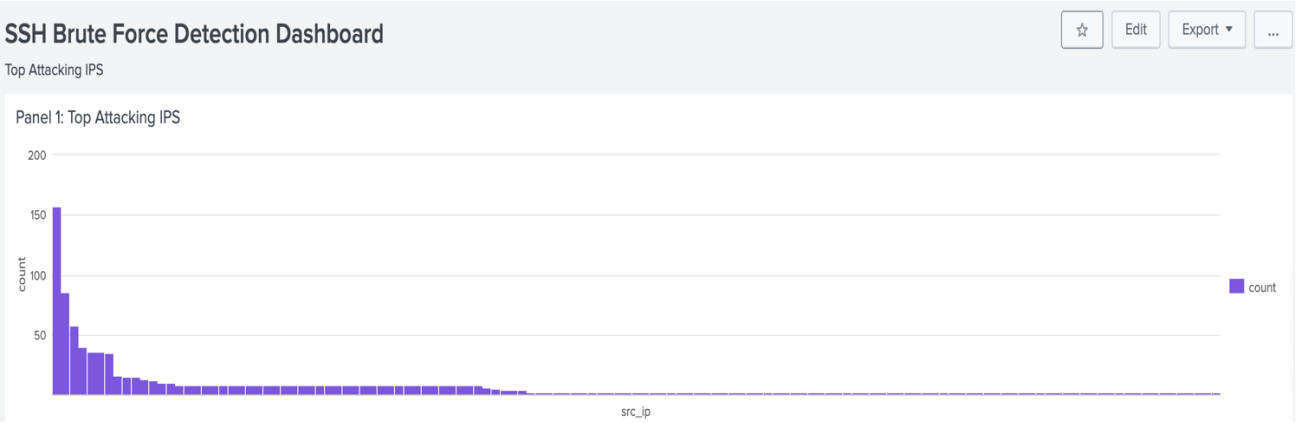
- Implement account lockout policies to prevent brute-force attacks
- Use multi-factor authentication (MFA) for SSH access
- Continuously monitor logs using SIEM tools like Splunk
- Set up alerts for repeated failed login attempts
- Block or blacklist high-risk IP addresses

**Skills Demonstrated**

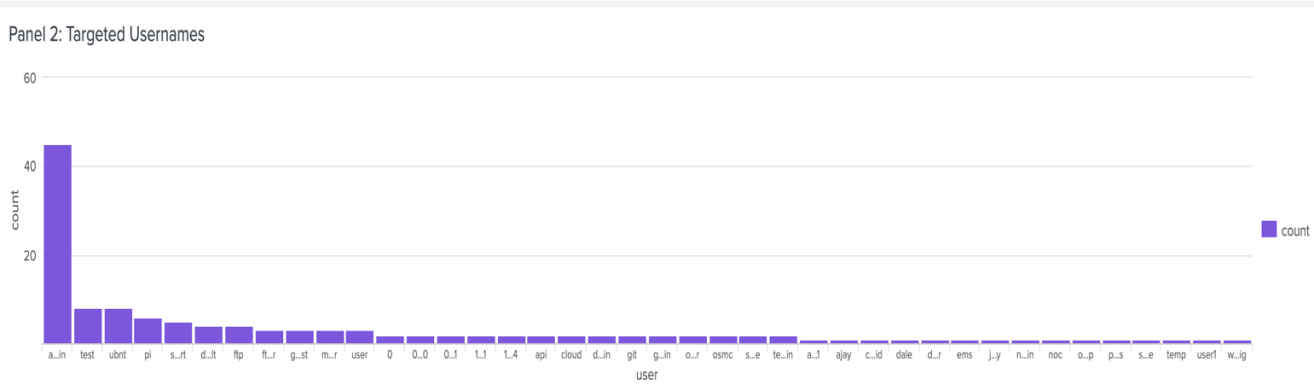
- Log ingestion and parsing in Splunk
- SPL query development
- Field extraction using regular expressions
- Cybersecurity threat detection
- Dashboard creation and data visualization
- Security analysis and reporting

## 📷 Screenshots

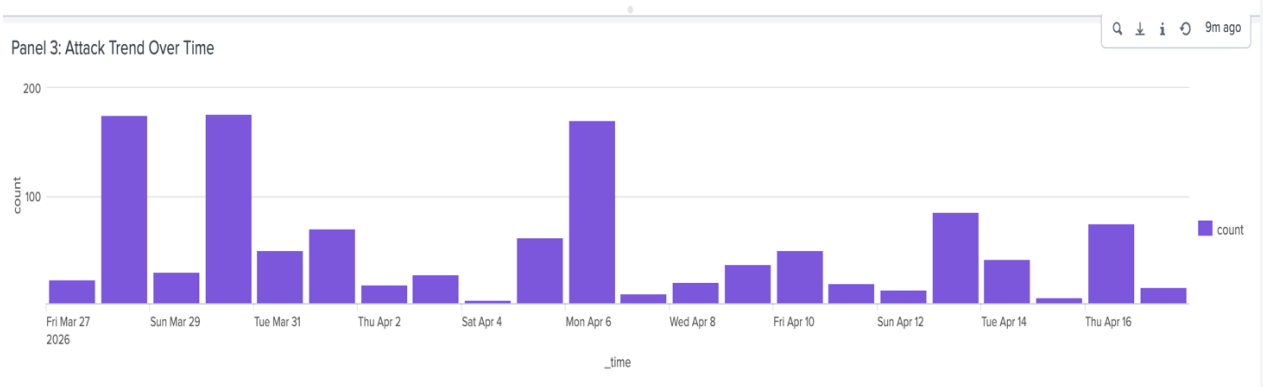
**PANEL 1:** TOP ATTACKING TRENDS




PANEL 2: TARGETED USERNAMES



PANEL :3 ATTACK TREND OVER TIME



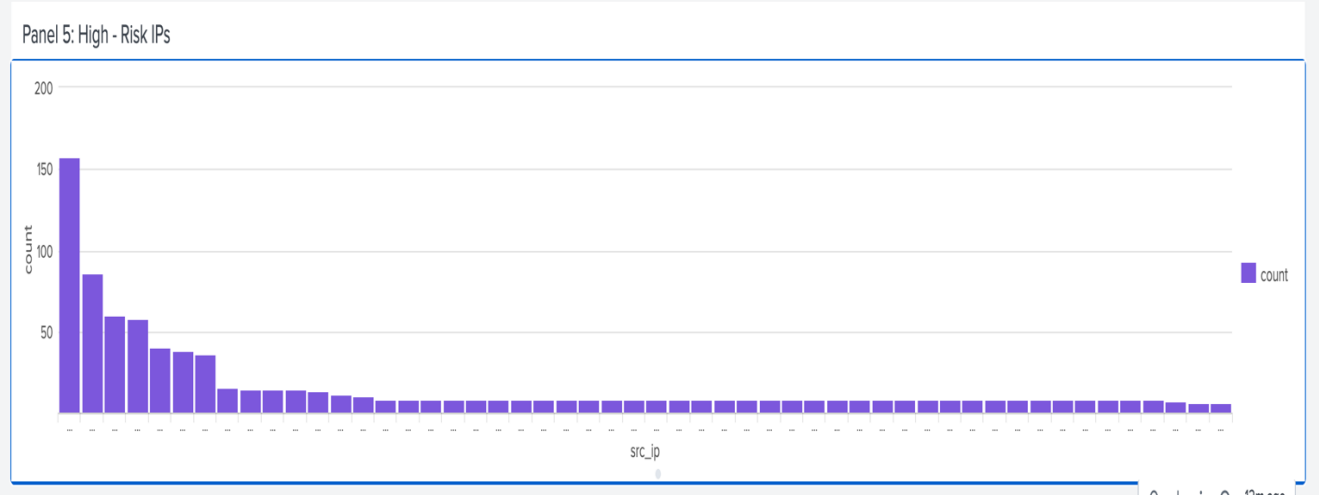

PANEL 4: SUCCESSFUL LOGINS

Panel:4 Successful Logins

| i | Event |
|---|---|
| > | Apr 17 14:54:07 ip-10-77-20-248 sshd[32491]: Accepted publickey for ubuntu from 85.245.107.41 port 55051 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr 20 14:14:29 ip-10-77-20-248 sshd[3964]: Accepted publickey for ubuntu from 85.245.107.41 port 51816 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr 20 14:13:36 ip-10-77-20-248 sshd[3819]: Accepted publickey for ubuntu from 85.245.107.41 port 51793 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr 10 13:09:29 ip-10-77-20-248 sshd[23655]: Accepted publickey for ubuntu from 85.245.107.41 port 55579 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr 10 10:53:51 ip-10-77-20-248 sshd[23132]: Accepted publickey for ubuntu from 85.245.107.41 port 52722 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr 10 10:18:00 server sshd[1242]: Accepted password for admin from 192.168.1.30 port 22 ssh2 |
| > | Apr 10 10:18:00 server sshd[1242]: Accepted password for admin from 192.168.1.30 port 22 ssh2 |
| > | Apr 10 10:18:00 server sshd[1242]: Accepted password for admin from 192.168.1.30 port 22 ssh2} |
| > | Apr 10 10:18:00 server sshd[1242]: Accepted password for admin from 192.168.1.30 port 22 ssh2} |
| > | Apr 10 10:18:00 server sshd[1242]: Accepted password for admin from 192.168.1.30 port 22 ssh2} |
| > | Apr 10 10:16:10 server sshd[1238]: Accepted password for user1 from 192.168.1.10 port 22 ssh2 |
| > | Apr 10 10:16:10 server sshd[1238]: Accepted password for user1 from 192.168.1.10 port 22 ssh2 |
| > | Apr 10 10:16:10 server sshd[1238]: Accepted password for user1 from 192.168.1.10 port 22 ssh2\ |
| > | Apr 10 10:16:10 server sshd[1238]: Accepted password for user1 from 192.168.1.10 port 22 ssh2\ |
| > | Apr 10 10:16:10 server sshd[1238]: Accepted password for user1 from 192.168.1.10 port 22 ssh2\ |
| > | Apr 10 10:10:00 ip-10-77-20-248 sshd[22761]: Accepted publickey for ubuntu from 85.245.107.41 port 52196 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr  9 18:30:15 ip-10-77-20-248 sshd[20660]: Accepted publickey for ubuntu from 95.93.96.191 port 64731 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr  7 17:25:19 ip-10-77-20-248 sshd[18054]: Accepted publickey for ubuntu from 85.245.107.41 port 55216 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr  5 11:13:44 ip-10-77-20-248 sshd[15109]: Accepted publickey for ubuntu from 85.245.107.41 port 49548 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |
| > | Apr  4 16:45:57 ip-10-77-20-248 sshd[8861]: Accepted publickey for ubuntu from 85.245.107.41 port 56363 ssh2: RSA SHA256:Kl8kPGZrTiz7g4FO1hyqHdsSBBb5Fge6NWOobN03XJg |

« Prev 1 2 3 4 5 6 7 8 9 10 Next »

PANEL 5: HIGH-RISK IPs




**Project Structure**

- /queries – SPL queries used
- /screenshots – Dashboard visuals
- /data – Sample log file
- /reports – Full project report

**Author**

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